Mathematical and Computational Methods

Exercise Sheet 3: Complex numbers II: exponential form, De Moivre, roots

Tutorial route (▶, in this order, about 80 minutes): 1.3, 2.4, 2.5(a,b), 2.6, 2.8, 2.10(a,b), 2.3. Everything else is for home study.

Levels: ⋆ drill, ⋆⋆ core, ⋆⋆⋆ challenge. Give exact answers unless a decimal is asked for.

1 Warm-up drills

Exercise 1.1 *(⋆ drill — exponential form).* Write each number in exponential form $r\,\mathrm{e}^{\mathrm{i}\theta}$ with $r > 0$ and $-\pi < \theta \le \pi$:

$$\text{(a) } 1+\mathrm{i}, \quad \text{(b) } -3, \quad \text{(c) } 2\mathrm{i}, \quad \text{(d) } -1-\mathrm{i}, \quad \text{(e) } \sqrt{3}-\mathrm{i}, \quad \text{(f) } -2+2\sqrt{3}\,\mathrm{i}.$$

Exercise 1.2 *(⋆ drill — back to Cartesian; Euler's identity).* Write each in Cartesian form $a + \mathrm{i}b$:

$$\text{(a) } 6\,\mathrm{e}^{\mathrm{i}\pi/6}, \quad \text{(b) } 3\,\mathrm{e}^{\mathrm{i}\pi}, \quad \text{(c) } \sqrt{2}\,\mathrm{e}^{-\mathrm{i}\pi/4}, \quad \text{(d) } 4\,\mathrm{e}^{\mathrm{i}3\pi/2}, \quad \text{(e) } \mathrm{e}^{\mathrm{i}5\pi/6}, \quad \text{(f) } \mathrm{e}^{\mathrm{i}2\pi}.$$

(g) Use (b) to state Euler's identity.

Exercise 1.3 *(▶ ⋆ drill — rotation).* Let $z = 2\,\mathrm{e}^{\mathrm{i}\pi/6}$. Write down $\mathrm{e}^{\mathrm{i}\pi/2}z$ in exponential form, and describe geometrically the effect of multiplying a complex number by (i) i, (ii) $\mathrm{e}^{\mathrm{i}\pi}$, (iii) $\mathrm{e}^{-\mathrm{i}\pi/2}$.

Exercise 1.4 *(⋆ drill — integer powers).* Compute, using the exponential form:

$$\text{(a) } (1+\mathrm{i})^4, \quad \text{(b) } (\sqrt{3}+\mathrm{i})^3, \quad \text{(c) } (1-\mathrm{i})^6, \quad \text{(d) } \left(2\,\mathrm{e}^{\mathrm{i}\pi/5}\right)^{10}.$$

(e) Use De Moivre's theorem to simplify $(\cos\theta + \mathrm{i}\sin\theta)^{-2}$.

Exercise 1.5 *(⋆ drill — a few roots).* Find all the (a) square roots of i and (b) fourth roots of 1. Plot the roots in (b) on an Argand diagram, mark the unit circle, and name the polygon they form.

Exercise 1.6 *(⋆ drill — the fundamental theorem of algebra).* (a) State the fundamental theorem of algebra, and say what "counted with multiplicity" means. (b) How many roots in $\mathbb{C}$, counted with multiplicity, has (i) a degree-5 polynomial; (ii) $z^7 - 3 = 0$; (iii) $(z-1)^2(z^2+1) = 0$? (c) True or false: every real cubic has at least one real root?

Exercise 1.7 *(⋆ drill — hyperbolic link).* Using $\cos\theta = \tfrac{1}{2}(\mathrm{e}^{\mathrm{i}\theta} + \mathrm{e}^{-\mathrm{i}\theta})$, evaluate $\cos(\mathrm{i})$ in terms of e. *(The general identity $\cos(\mathrm{i}x) = \cosh x$ is built up in Exercise 2.9.)*

Exercise 1.8 *(⋆ drill — phasors).* (a) Write $5\cos(\omega t + \tfrac{\pi}{3})$ and $2\sin\omega t$ as phasors $P = A\,\mathrm{e}^{\mathrm{i}\varphi}$. (b) The phasor $3\,\mathrm{e}^{-\mathrm{i}\pi/2}$ represents which real signal $A\cos(\omega t + \varphi)$?

2 Tutorial problems

Exercise 2.1 *(⋆⋆ core — arithmetic in exponential form).* Let $z = \sqrt{3} + \mathrm{i}$ and $w = 1 + \mathrm{i}$. Write z and w in exponential form, then find zw and z/w in *both* exponential and Cartesian form.

Exercise 2.2 *(⋆⋆ core — rotation in action).* (a) Rotate $z = 3 + 4\mathrm{i}$ anticlockwise by $\frac{\pi}{2}$ about the origin, and then (separately) clockwise by $\frac{\pi}{2}$. (b) Rotate $2 + 2\mathrm{i}$ anticlockwise by $\frac{\pi}{4}$. Give exact Cartesian answers.

Exercise 2.3 *(▶ ⋆⋆ core — prove De Moivre (stretch)).* Prove by induction that $(\cos\theta + \mathrm{i}\sin\theta)^n = \cos n\theta + \mathrm{i}\sin n\theta$ for all integers $n \geq 1$. (Pure-maths.)

Exercise 2.4 *(▶ ⋆⋆ core — powers by De Moivre).* Compute (a) $(-1+\mathrm{i})^{10}$ and (b) $(\sqrt{3}-\mathrm{i})^{-6}$ using De Moivre's theorem, giving Cartesian answers.

Exercise 2.5 *(▶ ⋆⋆ core — roots, two shapes).* Find (a) the cube roots of $8\mathrm{i}$ and (b) the sixth roots of -64. In each case give exact Cartesian values and describe the polygon they form on the Argand diagram.

Exercise 2.6 *(▶ ⋆⋆ core — fifth roots of unity).* (a) Write the fifth roots of unity in exponential form and sketch them on the unit circle. (b) Explain why they sum to zero (and why the same is true of the nth roots of unity for every $n \geq 2$). (c) By taking real parts of that sum, find the exact value of $\cos\frac{2\pi}{5} + \cos\frac{4\pi}{5}$.

Exercise 2.7 *(⋆⋆ core — data-science flavour: twiddle factors).* The fast Fourier transform is built from the "twiddle factors" $W_N^k = \mathrm{e}^{-\mathrm{i}2\pi k/N}$. Take $N = 8$. (a) Write W_8^1 in Cartesian form. (b) Show that $W_8^{k+4} = -W_8^k$ for every integer k (one of the symmetries the FFT exploits). (c) Give the modulus and argument of W_8^3.

Exercise 2.8 *(▶ ⋆⋆ core — $\cos 4\theta$ and $\sin 4\theta$).* Using De Moivre and the binomial theorem, show that $\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$, and write $\sin 4\theta$ as $\sin\theta$ times a polynomial in $\cos\theta$.

Exercise 2.9 *(⋆⋆ core — trig–hyperbolic link).* (a) Show $\cos(\mathrm{i}x) = \cosh x$ and $\sin(\mathrm{i}x) = \mathrm{i}\sinh x$. (b) Hence evaluate $\cos(\mathrm{i}\ln 2)$ exactly.

Exercise 2.10 *(▶ ⋆⋆ core — combining oscillations (physics)).* (a) Combine $8\cos\omega t + 15\sin\omega t$ into a single sinusoid $R\cos(\omega t - \alpha)$. (b) Combine $2\cos\omega t + 2\cos(\omega t + \frac{\pi}{2})$ likewise.

3 Further practice (home)

Roots and the fundamental theorem

Exercise 3.1 *(⋆⋆ core — solving $w^n = z$ with a non-real right-hand side).* (a) Solve $w^3 = -27\mathrm{i}$, giving the three roots in exponential and Cartesian form. (b) Solve $w^3 = 1 + \mathrm{i}$, leaving the roots in exponential form, and check that each root is the previous one rotated through $\frac{2\pi}{3}$ about the origin.

Exercise 3.2 *(⋆⋆ core — square root the algebraic way).* The argument of $5 - 12\mathrm{i}$ is not a standard angle, so the exponential method is awkward. Instead find both square roots of $5 - 12\mathrm{i}$ in Cartesian form by equating real and imaginary parts of $(a + \mathrm{i}b)^2$, and verify by squaring.

Exercise 3.3 *(⋆⋆ core — FTA and real factors)*. (a) Verify that $z = 2\mathrm{i}$ is a root of $z^2 + 4$, and write down the other. (b) Factor $z^4 + 4$ into two real quadratics. [Hint: find the four roots first.]

Exercise 3.4 *(⋆ ⋆ ⋆ challenge — a conjugate equation)*. Find all $z \in \mathbb{C}$ with $z^2 = \overline{z}$.

Trigonometric identities

Exercise 3.5 *(⋆⋆ core — power reduction)*. Use the inverse Euler formula to write $\sin^4\theta$ as a sum of cosines of multiples of θ.

Exercise 3.6 *(⋆ ⋆ ⋆ challenge — a finite cosine sum)*. Using the geometric series for $\sum_{k=0}^{n-1} \mathrm{e}^{\mathrm{i}k\theta}$ (with $\theta \neq 0 \bmod 2\pi$), show that

$$\sum_{k=0}^{n-1} \cos k\theta = \frac{\sin(n\theta/2)}{\sin(\theta/2)} \cos\!\left(\tfrac{(n-1)\theta}{2}\right).$$

Trigonometric–hyperbolic link

Exercise 3.7 *(⋆⋆ core — the converse pair)*. Show that $\cosh(\mathrm{i}x) = \cos x$ and $\sinh(\mathrm{i}x) = \mathrm{i}\sin x$, and evaluate $\sin(2\mathrm{i})$ and $\cos(\pi\mathrm{i})$ in terms of e.

Phasors

Exercise 3.8 *(⋆⋆ core — physics word problem: two AC sources)*. Two alternating voltages of equal amplitude but different phase, $V_1 = 10\cos\omega t$ and $V_2 = 10\cos(\omega t + \frac{2\pi}{3})$ (volts), are added. Find the amplitude and phase of $V_1 + V_2$.